2026 International Seminar on Intelligent Business and Edge-Computing Research (ISIBER) | 979-8-3315-5891-8/26/$31.00 ©2026 IEEE | DOI: 10.1109/ISIBER68248.2026.11470090

Performance Analysis of Time Series Models to Predict Sales Activity of SMEs

1st Justine Christian Susanto
Entrepreneurship Department
BINUS Business School
Bina Nusantara University
Bandung, Indonesia
justine.susanto@binus.ac.id

2nd Maureen Letitia Wiratama
Entrepreneurship Department
BINUS Business School
Bina Nusantara University
Bandung, Indonesia
maureen.wiratama@binus.ac.id

3rd Patrick Christian Nathaniel
Entrepreneurship Department
BINUS Business School
Bina Nusantara University
Bandung, Indonesia
patrick.nathaniel@binus.ac.id

4th Ivan Diryana Sudirman
Entrepreneurship Department
BINUS Business School
Bina Nusantara University
Bandung, Indonesia
ivan.diryana@binus.ac.id

***Abstract*—Micro, Small, and Medium Enterprises (MSMEs) are a key pillar of the Indonesian economy. Many small businesses implement pre-order systems to minimize inventory risk and high initial capital requirements. However, pre-order systems also pose unique challenges, particularly in terms of uncertain production costs and potential delays in fulfilling orders. Daily sales in these systems tend to be unstable, with periods of no transactions and sporadic spikes in demand. The purpose of this study is to develop and evaluate a forecasting model for MSME data operating with a pre-order system and to test whether machine learning tools such as XGBoost can produce reliable performance. This pattern results in a high frequency of zero values, making accurate predictions difficult. To predict such sales patterns, this study tests a Predictive Regression Model using the XGBoost technique and compares it with a baseline method. The data used consists of daily sales from 2023 to 2025. The data is then augmented with lag features and rolling window-based statistical features.**

***Index Terms*—Machine learning forecasting, XGBoost regression, SMEs, demand forecasting,**

I. Introduction

Micro, small, and medium enterprises (MSMEs) play a significant role in the Indonesian economy [1]. Newly established businesses generally face limitations in capital, human resources (HR), and production capacity. In situations like this, many MSMEs choose a pre-order system as their primary strategy to mitigate operational risk. This sales system involves producing new goods after confirmation and agreement on the order. This system not only opens up business opportunities but also reduces stockpiling, reduces idle raw materials, and increases inventory turnover. This allows new entrepreneurs to experiment without incurring significant financial risks.

On the other hand, while this pre-order system can reduce inventory-related risks, it also poses other equally significant risks. Because customers have paid in advance and have high expectations regarding the arrival time of their orders, when demand increases or capacity is insufficient, the risk of late and inaccurate delivery increases, directly impacting customer confidence. Furthermore, sales data in the PO system tends to fluctuate. Sometimes there are days without orders, followed by several days of surges in demand. These patterns lead to volatile sales activity and make it nearly impossible to predict based solely on daily trends and standards.

The disorganized data structure shows that even though the Pre-Order (PO) system itself does not require inventory control, the business still requires stock control for other purposes [2]. Forecasting is an important foundation in this context, because it can help in predicting trends in workload as well as providing production capacity. Apart from that, forecasting is also valuable for dealing with human resource needs and predicting opportunities for delays. By having an understanding of potential future activities, industry can take preventive action such as temporarily increasing its workforce, increasing the purchase of certain raw materials and changing production processes to prepare itself for variations in demand.

The objectives of this study include the development and evaluation of a forecasting model for sales data from SMEs operating under an Pre-Order (PO) system, taking into account the characteristics of the SME data. In addition, there is also an additional objective, namely conducting an analysis to determine whether machine learning models such as XGboost can provide better results compared to basic methods, such as naive forecasting and moving averages. This study also attempts to provide an overview of how forecasting works to help not only increase inventory, but also as a means to improve operational preparation and prevent risks in terms of order fulfillment in the PO system.

II. Related Work

Research confirms that studies of pre-order sales systems point to the fact that pre-order data alone is not a sufficient tool to eliminate the need for forecasting. Instead, it increases the need for much more sophisticated and complex predictions. In the publishing industry, pre-order data, while providing some insight into potential demand, is still insufficient to meet all the needs of identifying post-launch product dynamics, understanding seasonal demand patterns, and understanding the impact of specific marketing activities [3]. In the area of procurement and inventory management, research has shown that data-driven systems require organizations to combine preliminary order data with algorithms used to predict demand. This aims to consider factors such as uncertainty in the conversion process from purchase orders to formal orders, the likelihood of cancellations, and the need for restocking or replenishment of inventory [4], [5].

Research conducted by Sampedro [**?**], shows that the frequency level of messages before sales or what is known as pre-orders, is a dynamic and very significant phenomenon that requires continuous prediction through statistical tools such as time series methodology and ARIMA (AutoRegressive Integrated Moving Average) models. With these results, it can be concluded that the pre-order system, in general, adds a new path in the forecasting process, namely through estimates of the direction of message growth before recent sales are made. In e-commerce, computer-based predictive models are often used as a tool to combine information about previous orders with actual transactions and potential buyer behavior. Their primary goal is to predict the efficiency of the conversion process and the future demand for goods after a new product is released to the market [6]. In the fashion industry, a method known as a hybrid of artificial neural networks and logistic regression has demonstrated success in predicting pre-order numbers, consumer interest, and additional demand directly or indirectly related to those orders during rapidly changing market trends [7].

AHAR 24's food ordering system management implements a pre-order system with a forecasting model to manage raw material requirements, customer capacity, and staff scheduling. [8]. Various studies have shown that pre-order systems are very useful, especially when used in the early stages. However, these systems still require a predictive model to manage potential unavailability of demand, operational capacity, and risk to maintain customer satisfaction.

Traditional forecasting methodologies tend to be unable to recognize rapidly changing data and dynamic market changes. This can result in significant operational risks, such as overstocking or understocking. Research has shown that analytics-based predictions using machine learning have the potential to provide more accurate demand forecasting results. This approach also takes into account various factors, such as seasonal patterns, various external variables, and operational data that can be collected in real time. [9].

XGBoost is a machine learning tool that can be used for prediction and is frequently used for this purpose, particularly in data facing Zero Inflation, where the data contains many zero values. XGBoost addresses imbalanced distributions, patterns of falls, or relatively rare events. These are typically difficult for standard statistical models to handle. For example, in the insurance industry, the XGBoost algorithm has been used to build models that can predict the frequency of vehicle claims, where a large percentage of consumers do not file a claim during a given period. XGBoost can effectively learn patterns from this type of structured data and consistently provides more accurate predictions than traditional methods for capturing rare events. [10], [11].

III. Methodology

This research involved several steps to develop a predictive model based on daily transactions. These steps were then implemented in stages, starting with data integration, feature development, model training, and then comparing it with the baseline approach.

The first step is to collect the data and then clean it. This process involves combining multiple data cells into a single structure. The data format is then normalized and checked to ensure data integrity. The data set is then reindexed into a complete daily sequence. Furthermore, empty values on the daily data sets due to no transactions are filled with 0s to ensure consistent daily frequency.

The next stage involved developing features for the model that describe time series patterns. Time-based variables such as date, month, year, and day of the week were incorporated into the system. Lag features (1, 2, 3, 7, 14, 21, and 30 days) and rolling window statistics were also developed, including the mean, standard deviation, maximum, and minimum values for 3, 6, 14, and 30-day windows. Other features introduced included indicators for days without transactions, their ratios over short and medium time periods, and spike indicators based on transaction value percentiles. These features provided a more predictive structure for the machine learning model.

The third stage involves developing the predictive objective and separating the dataset. The prediction objective is the sales value for the following day, namely day t+1. In this case, the dataset is separated using a time-based split method, where some initial data is allocated for the training process, while the remainder is used for the testing phase. The primary purpose of this separation step is to ensure that no information leaks or creeps between the two periods.

In the fifth stage, the forecasting model development process continues. The XGBoost regressing algorithm was chosen for the main model. This model was embedded or trained using all the predefined features. In addition to this model, two more basic approaches were developed: a naive forecast, which relies on the previous day's value for forecasting, and a seven-day objective moving average, which is a deliberately simple smoothing method. The purpose of these two approaches is to provide an objective assessment of the main model's performance.

The evaluation is carried out using various common metrics commonly used in time series data analysis, such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) for periods where the observed values are non-zero. Additionally, Weighted Average Percentage Error (WAPE) can be used as another measure. To understand the model's behavior in more detail and over time, a visualization summarizing the comparison of actual values with predicted results during the test period has been prepared and is ready for use.

The final stage involves compiling the findings and analyzing their impact, including describing data patterns, relationships between used features, model results, and the operational environment within the pre-order system. This stage also includes directions for further development and the potential use of other approaches, such as classification methods, to predict order volume or potential fluctuations.

IV. Results and Discussion

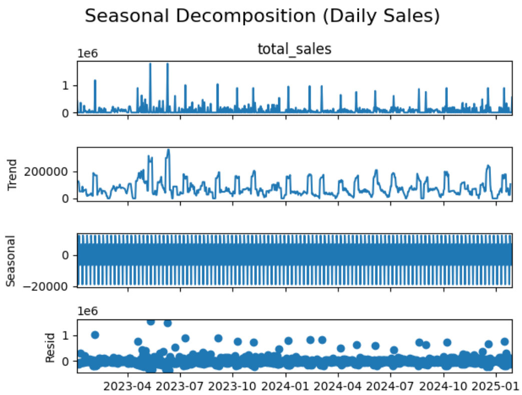

Fig. 1. Seasonal Decomposition (Daily Sales)

Figure 1 shows the results of the seasonal decomposition of the total sales variable. The trend graph outlines the general movement of daily sales. The seasonal panel shows a fairly strong repeating pattern. The residual panel shows daily variations not accounted for by the other components.

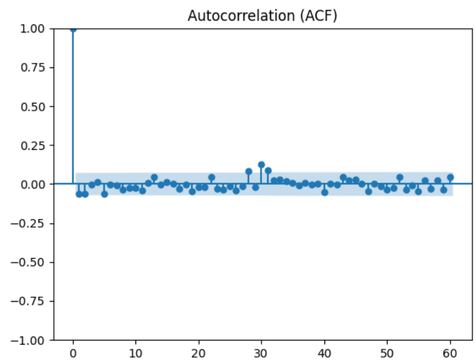

Fig. 2. Autocorrelation Function (ACF)

Figures 2 and 3 show the ACF and PACF plots. Both graphs show that the short-term autolag correlation is relatively small, except for certain lags where it appears slightly prominent.

To illustrate the distribution of days without transactions, the figure 4 shows a comparison between the number of days with zero sales and the number of days with positive transactions. This data shows that the number of days without transactions is indeed quite dominant.

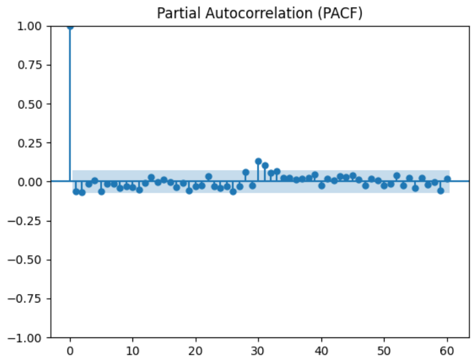

Fig. 3. Partial Autocorrelation Function (PACF)

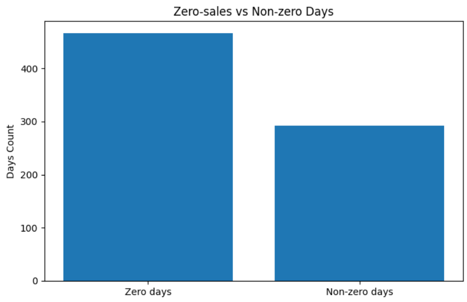

Fig. 4. Zero-sales vs Non-zero Days

The sales pattern based on day of week and month is shown in Figure 5 in the form of a heatmap of average daily sales.

A cross-year comparison is shown in Figure **??**, which plots the time series by calendar year to see temporal changes between periods.

Figure 7 shows the 7-day and 30-day moving averages, which serve as a stabilizer for short-term fluctuations.

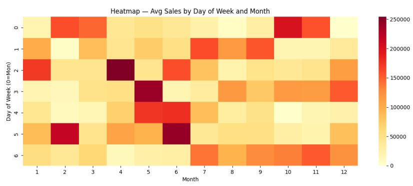

Fig. 5. Heatmap — Average Sales by Day of Week and Month

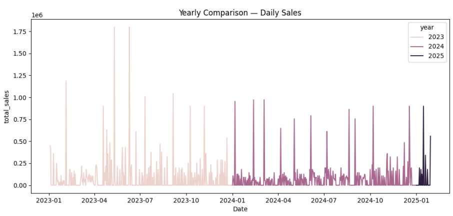

Fig. 6. Yearly Comparison — Daily Sales

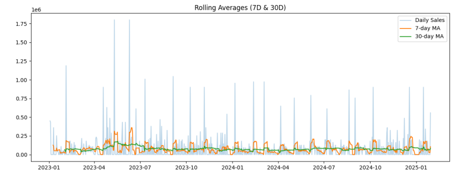

Fig. 7. Rolling Averages (7D & 30D)

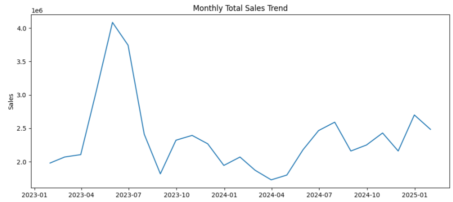

Fig. 8. Monthly Total Sales Trend

Finally, total monthly sales are visualized in Figure 8 to see the aggregate dynamics at the monthly level.

The feature dataset contains 36 variables, including daily calendars, lag features, rolling statistics (3, 7, 14, and 30 days), and a spike indicator (is_spike), the dataset spans 727 days, with 581 train days and 146 test days.

TABLE I
PERFORMANCE COMPARISON (TEST SET)

Model	**MAE**	**RMSE**	**MAPE**	**WAPE**
XGBoost	116.350	185.854	64,35%	153,45%
Naive	125.014	225.411	87,02%	164,88%
MA7	103.949	170.920	54,20%	137,10%

The XGBoost model was trained on all features, and the test results are shown in Table I. The baseline models used were Naive Forecast and a 7-day Moving Average (MA7).

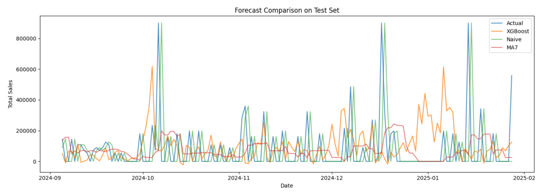

Fig. 9. Forecast Comparison on Test Set

Figure 9 shows a comparison between the actual values and the three prediction methods over the test period.

A. Discussion

The results of the exploration in the initial phase showed that the daily sales pattern had very unstable characteristics. The visualizations in figures 1 to figures 8 show a combination of days without transactions, spikes that occur on certain days, and a lack of significant seasonal patterns. This differs from sales patterns in conventional retail markets, which generally have a clear weekly or monthly rhythm. The daily dominance of zeros, as seen in Figures 4 and 7, indicates that my business process is more akin to a pre-order system than a regular stock system. In patterns like these, sales don't occur naturally but are influenced by orders coming in on a given day.

This irregularity is also evident in the ACF (AutoCorrelation Function) and PACF (Partial AutoCorrelation Function). Both show a lack of consistent autocorrelation or very low consistency. This indicates that sales activity on one day cannot easily be used as a basis for estimating or predicting sales results on the following day. However, the strongest signals are obtained from activity at much longer lag intervals or ranges.

The model performance shown in Table I reinforces this point. XGBoost was unable to outperform MA7 and only slightly outperformed the baseline method. The MAEs and RMSEs of all models were generally quite large compared to the actual sales volume. The high MAPE values are also understandable, as there are many days with zero sales. When actual zeros are prevalent, even small absolute errors can lead to significant inconsistencies. This makes metrics like MAE or MAPE less suitable for measuring performance on datasets with a large number of zeros, although they can still provide a general overview of performance.

Viewing the data phenomenon from this perspective, MA7 shows more positive results than other methods. This is because the MA7 method is able to reduce variations that occur over short time periods and is better suited to processing data containing many extreme fluctuations. While MA7 cannot be categorized as a particularly intelligent or intelligent model, simple smoothing techniques are generally a strong initial choice when the data does not exhibit clear autocorrelation. Alternatively, XGBoost requires a more consistent pattern to effectively utilize features like lag and rolling. However, if the daily pattern in a dataset is inconsistent and the majority of activity is influenced by sporadic orders, then a model based on tree-based boosting techniques will lose its structuralism, which is its main advantage.

The prediction comparison visualization in Figure 9 demonstrates this more concretely. All three methods repeatedly failed to track sudden sales peaks. Nearly all significant spikes fell outside the predictable region of statistical patterns. In the context of PO data like this, spikes often stem from a single large order batch, a specific promotion, or an operational decision not visible from daily sales data. In other words, the spikes are exogenous to the available patterns.

Despite this, the modeling process still yielded several valuable findings. First, the 30-day rolling variable and the lag of more than one week remain significant features in the model, indicating that despite the volatility it has a predictable medium-term pattern. Furthermore, the high frequency of zero sales confirms that forecasting is not easy. (as shown in Figures 4 and 7, However, it is also a tool for better understanding the potential risks of zero-transaction activity. This is especially true in situations where the ordering system is operating at maximum production capacity, requiring planning well in advance.

The results of this study indicate that time-series methods, in general, are not very effective for business text content that relies heavily on pre-order systems. This is especially true when the increase in orders does not immediately constitute a clear historical pattern. Nevertheless, this study provides further insight into the data structure of the volatility evaluation level. Furthermore, it identifies risky periods in the production process and also the risk of exceeding certain delivery levels.

B. Managerial Impact

The results of this study provide several management implications for micro, small, and medium enterprises operating using a pre-order system. This system creates unstable daily sales patterns and is largely influenced by sporadic spikes in demand. Nevertheless, the modeling results can provide valuable information for operational planning.

The comparative results of the prediction models show that the MA7 method, although simple, can provide relatively more stable estimated values compared to other methods. In the context of managerial systems, this means that MSMEs can benefit from short-term moving averages as a basic indicator for forecasting the basic value of primary demand. This indicator can be a useful tool in the process of mapping production capacity on certain days, especially in efforts to ensure that human resources and raw materials are always in optimal readiness or prepared, so they can effectively handle potential increases in demand without over-preparation.

While the model is not yet capable of highly accurate predictions regarding production spikes, identifying day-to-day periods of extremely high production levels still provides a good overview of potential operational risk points. Changes in production volumes. Significant changes, especially sudden ones, can lead to production delays. This significantly impacts customers because they have already paid before receiving the product. Therefore, customers have clear expectations regarding product timing and quality. This information forms the basis for planning reserve capacity, whether through additional staffing, shift scheduling, or other planning alternatives.

Several features deemed important by the model, namely the 40-Day Lag, 20-Day Lag, and 30-Day Moving Average, indicate that medium-term patterns play a significant role in demand prediction. These findings can be utilized by management to plan production by considering the periodic rhythm of product demand. Furthermore, these results demonstrate that pre-order activity is not entirely random, and therefore can be leveraged to improve the accuracy of raw material availability in inventory management and reduce commonly occurring adjustment costs.

V. Conclusion

The results of his research showed that the sales pattern of SMEs based on the pre-order system tends to be unstable, as seen by several days a week with no transactions at all, as well as irregular or sporadic spikes in orders. Using a regression-based predictive model like XGBoost, it turned out to be difficult for the method to consistently capture the dynamics of these sales patterns. However, other simple methods such as the Moving Average (MA) with a 7-day period showed the most stable and reliable results as the main basis for initial planning. However, its accuracy is still limited due to the sparse and discontinuous nature of the data. This is a major limitation of this study, as the daily regression approach is not a perfect solution when the data structure exhibits more of a lag pattern than a continuous pattern. Consequently, future research should consider classification methods, such as designing models that model the likelihood of orders occurring or the peak probability on a given day, which are considered more appropriate to the PO operational style.

References

[1] N. Aprilia, W. T. Subroto, and N. C. Sakti, "The role of small and medium enterprises (smes) in supporting the people's economy in indonesia," *International Journal of Research and Scientific Innovation*, vol. XI, no. XII, pp. 368–376, 2025.

[2] V. Chandan, S. H. Badugu, N. Teja, A. V. N. S. Harsha, D. Devendra, M. A. M. Umair, and E. C. Jones, "Optimization of production and inventory control," *International Supply Chain Technology Journal*, vol. 8, no. 1, 2022.

[3] R. Pizzano, "Sales forecasting of products with very short life cycles," *Journal of Business Forecasting*, vol. 30, no. 2, p. 4, 2011, retrieved from Questia database. [Online]. Available: https://www.questia.com/library/journal/1P3-2466731421/sales-forecasting-of-products-with-very-short-life

[4] X. Xiong, Y. Li, W.-D. Yang, and H. Shen, "Data-driven robust dual-sourcing inventory management under purchase price and demand uncertainties," *Transportation Research Part E: Logistics and Transportation Review*, vol. 160, p. 102671, 2022.

[5] L. Xu, X. Liu, L. Xiao, Q. Zuo, J. Liu, and W. K. V. Chan, "Data-driven procurement optimization in fresh food distribution market under demand uncertainty: A two-stage stochastic programming approach," in *Proceedings of the IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 2022, pp. 869–875.

[6] B. Bedir, "Machine learning based demand forecast models for e-commerce industry," 2022, retrieved from ResearchGate. [Online]. Available: https://www.researchgate.net/profile/Bulent-Bedir/publication/365375831

[7] R. Zahabiyah and A. Cakravastia, "A hybrid artificial neural network and logistics regression model for fashion sales strategies prediction," in *Proceedings of the International Conference on Industrial Engineering and Operations Management*. Monterrey, Mexico: IEOM Society, Nov. 2021, pp. —, november 3–5, 2021. [Online]. Available: http://ieomsociety.org/proceedings/2021monterrey/377.pdf

[8] S. B. Iqbal, "Ahar24: A modern restaurant food delivery management system," Project Report, Department of Computer Science and Engineering, BRAC University, Nov. 2024, m.Sc. in Computer Science and Engineering. [Online]. Available: http://hdl.handle.net/10361/25401

[9] A.-F. A. Adetula and T. Akanbi, "Beyond guesswork: Leveraging ai-driven predictive analytics for enhanced demand forecasting and inventory optimization in sme supply chains," *International Journal of Science and Research Archive*, vol. 10, no. 2, pp. 1389–1406, 2023.

[10] B. So, "Enhanced gradient boosting for zero-inflated insurance claims and comparative analysis of catboost, xgboost, and lightgbm," *Scandinavian Actuarial Journal*, vol. 2024, no. 10, pp. 1013–1035, 2024.

[11] N. Kollongei and F. Onyango, "Motor insurance claim frequency prediction using xgboost," *Asian Journal of Probability and Statistics*, vol. 26, no. 10, pp. 155–170, 2024.